EELU
الجامعة المصرية للتعلم الإلكتروني الأهلية
THE EGYPTIAN E-LEARNING UNIVERSITY

Fall Semester

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Information Technology

IT438

Communication Technology

Kamal Hamza, PhD

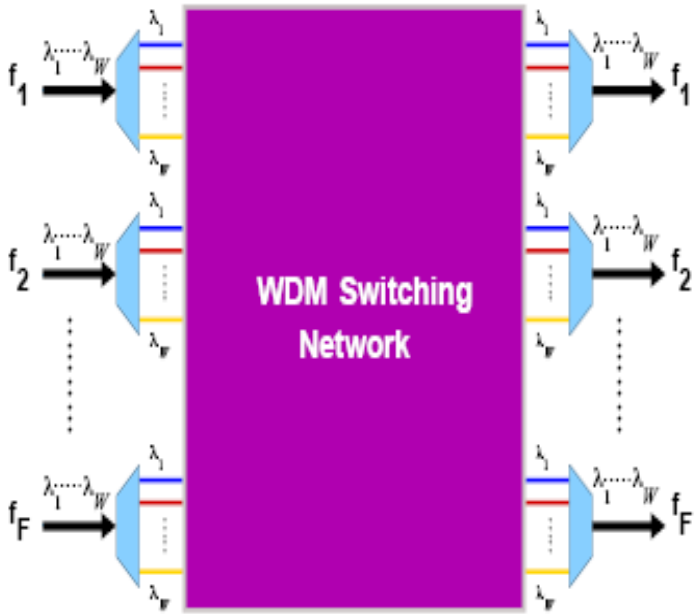
khamza@eelu.edu.eg

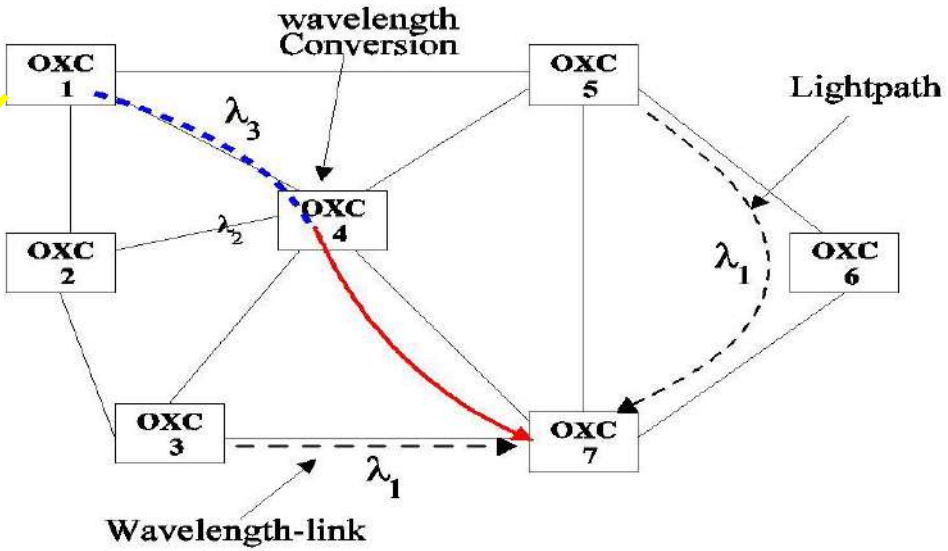
Acknowledgement: *This presentation contains some figures and text from Data Communications and Networks book by W. Stallings*

Network Generations

Network Generation	Network Type	Transmission Medium	Transmission Technology	Node Technology	Current Status
G1	Electronic Network	Copper cables	Electronic	Electronic	Still in use: e.g. access networks, LANs, and end-user telephone networks
G2	Optical Network	Optical fiber	Optical (single wavelength)	Electronic	Still in use for small-scale optical systems and LANs.
G3	Optical Network	Optical fiber	WDM	Electronic and Hybrid	Currently deployed in backbone networks (WANs), MANs, and LANs.
G4	(All) Optical Network	Optical fiber	WDM	Optical	In research Laboratories

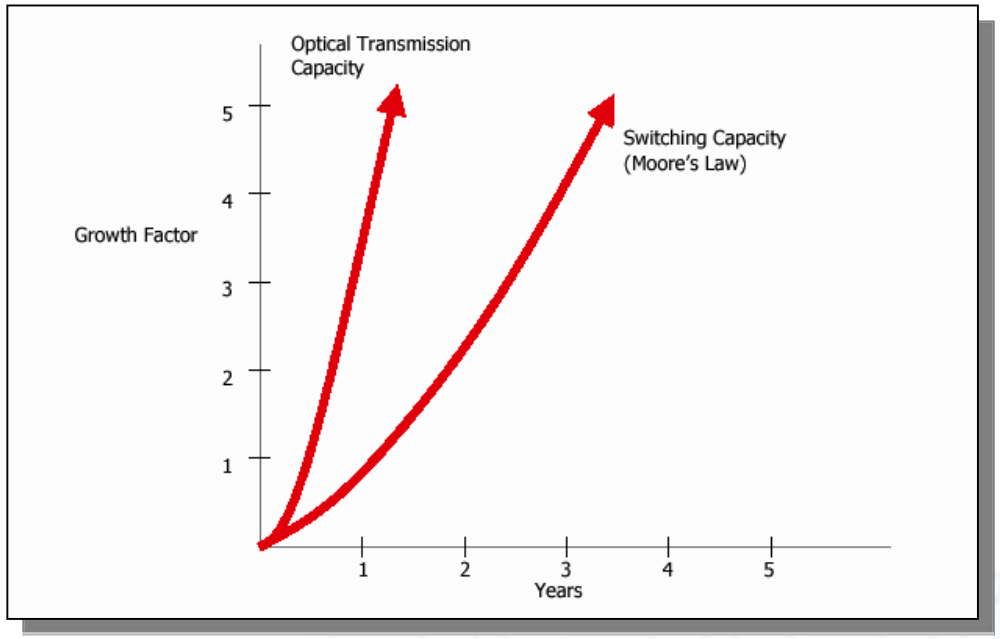
All Optical Network



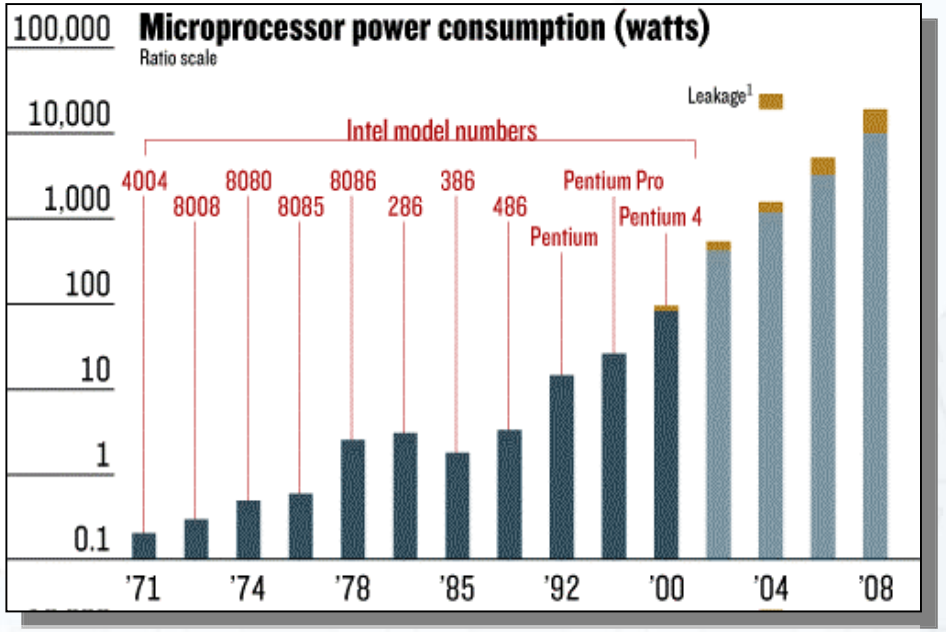


All-Optical Networks: Needs and Challenges

Fiber/Processor Bottlenecks



Fiber Capacity Increase Outstrips Electronic Switching Capacity Increase



Microprocessors will Dissipate Increasing Power with Today's Technology

WDM Networks Evolution

WDM Network Evolution

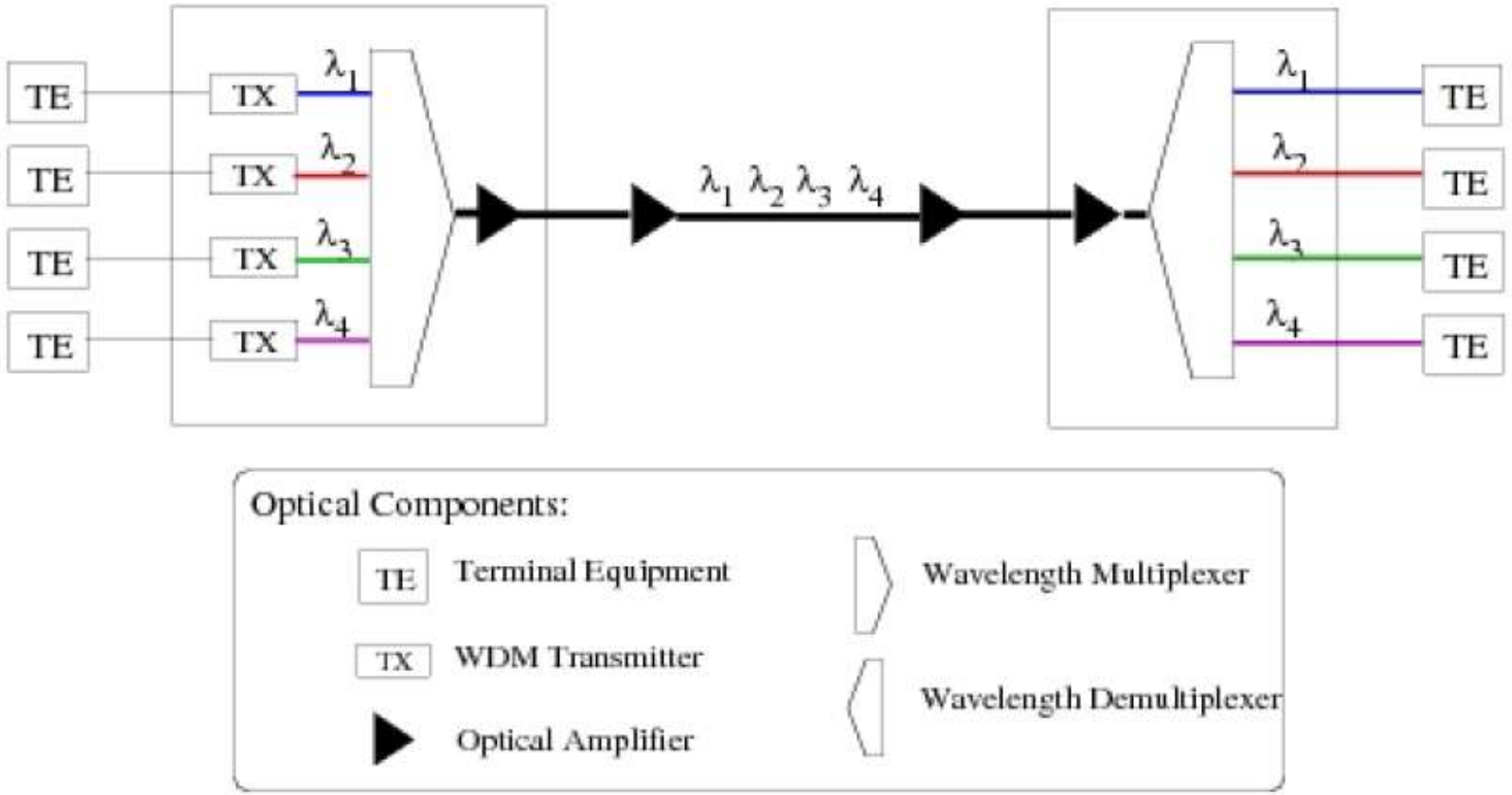
- The three main generations of WDM networks are:

  1- Point-to-Point WDM systems,

  2- Wavelength Add/Drop Multiplexer (WADM) or Optical Add/Drop Multiplexer (OADM), and

  3- Fiber and Wavelength Crossconnects (passive star, passive router, and active switch).

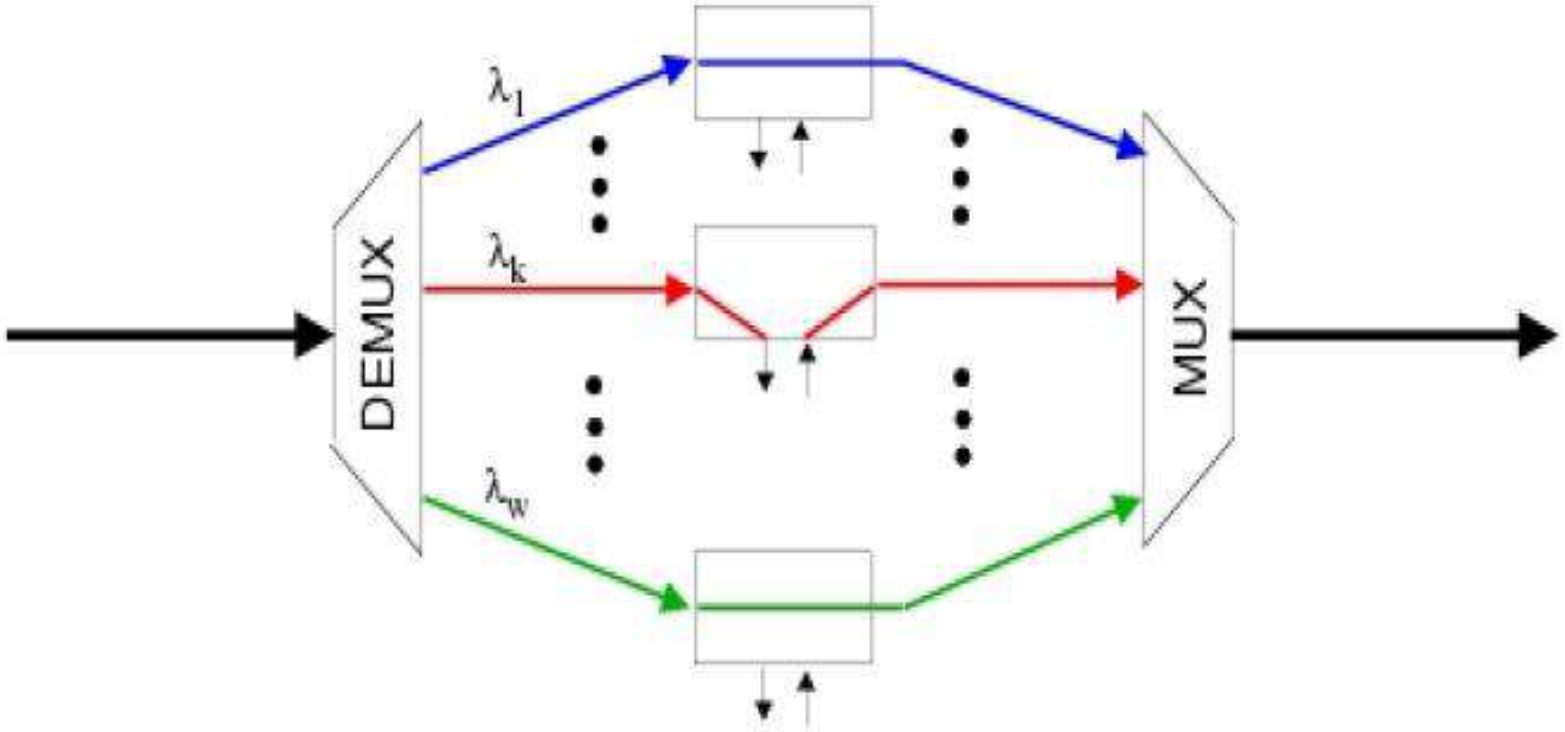
1- Point-to-Point WDM System



1- Point-to-Point WDM System

- The first generation of WDM networks (early to mid 90’s) was deployed as a cost-effective mean to upgrade the transmission capacity of point-to-point communication links.
- WDM point-to-point systems have been commercially available for more than a decade. Channel counts of 64 are common. The maximum channel count today is 160, and this number will reach 320 in the near future.

2- Wavelength Add/Drop Multiplexers



2- Wavelength Add/Drop Multiplexers (count.)

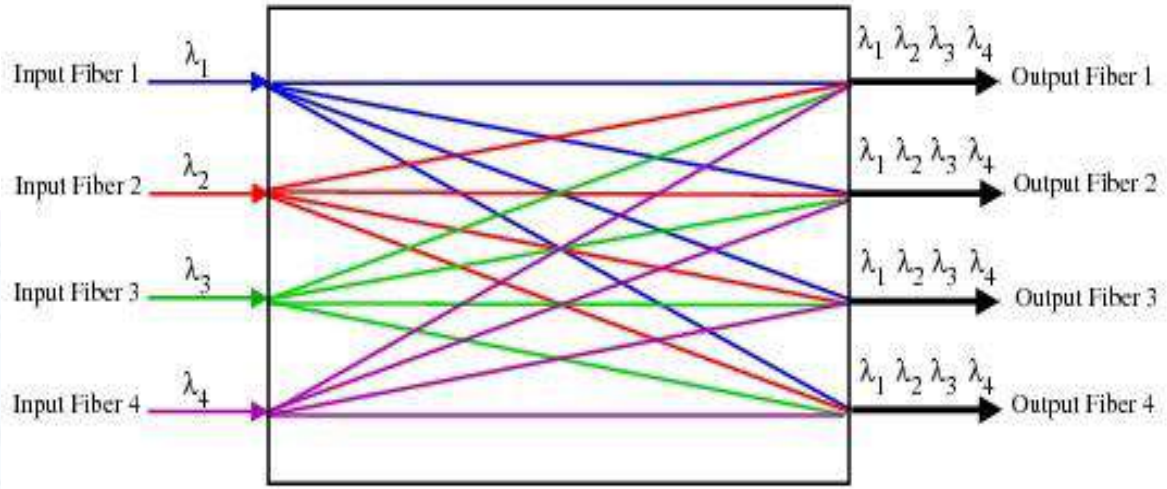
- A WADM consists of a DEMUX, followed by a set of switches , one per wavelength, followed by a MUX.
- WADM provides a great flexibility to WDM transmission links by allowing the capability of adding/dropping data streams at an intermediate points in the transmission link.

3- Fiber and Wavelength Cross-Connects

- The introduction of WADM increased the flexibility of the point-to-point WDM system, however, it does not provide the capability of interconnecting a set of multiwavelength fiber links, i.e., a true “WDM networking” infrastructure.
- The evolution of optical “interconnection” devices in the mid 90’s has marked the birth of the third generation of WDM systems — *WDM networks*.
- Based on the interconnection capability, we can distinguish three categorizes of optical interconnection devices:
  - Passive Star
  - Passive Router
  - Active Switch

Passive Star

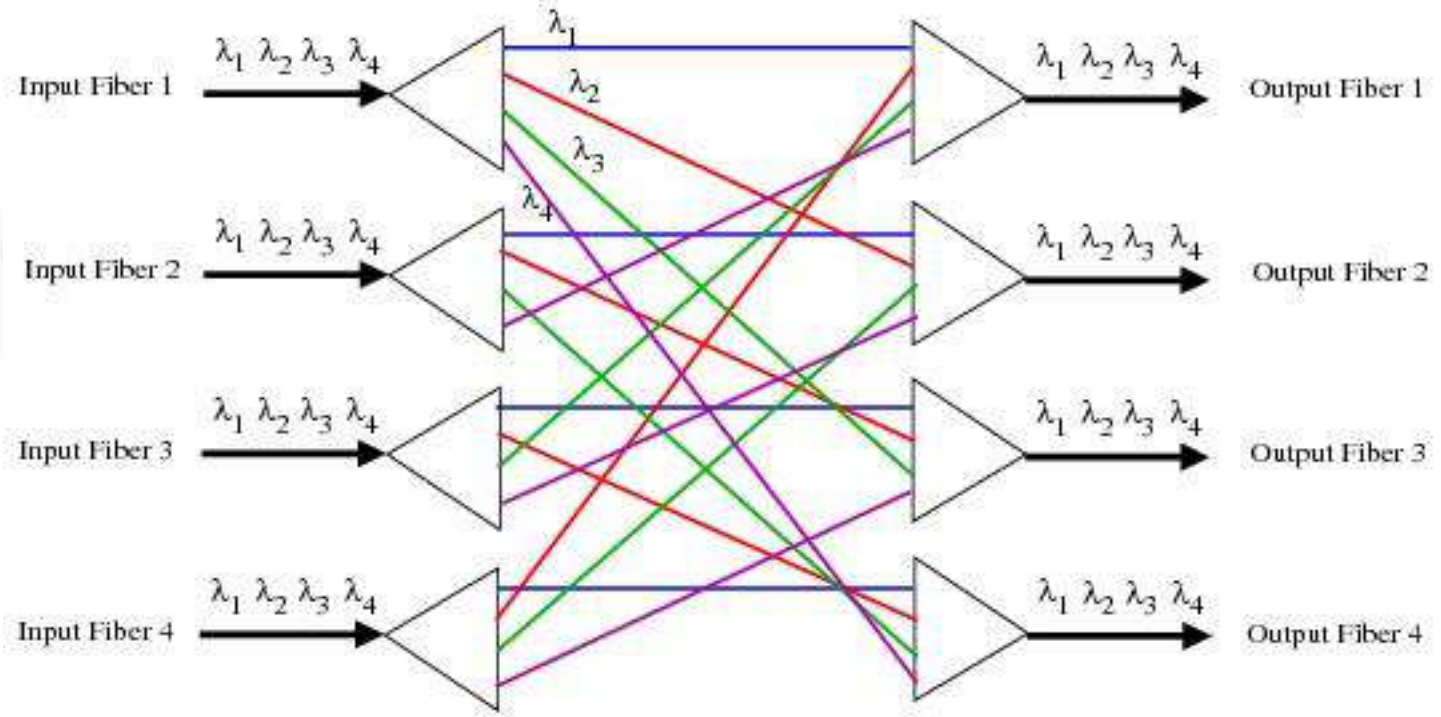
- The basic functionality of a passive star is as follows: the power of an input signal on a given wavelength from an input fiber port is equally split among all output fibers.



- Note that, in a passive star, a collision will occur when two or more signals from the input fibers are simultaneously launched into the star on the same wavelength.

Passive Router

- manage the routing of optical signals by utilizing predetermined paths or configurations. They can combine or split signals based on specific wavelengths without any active signal processing or amplification.



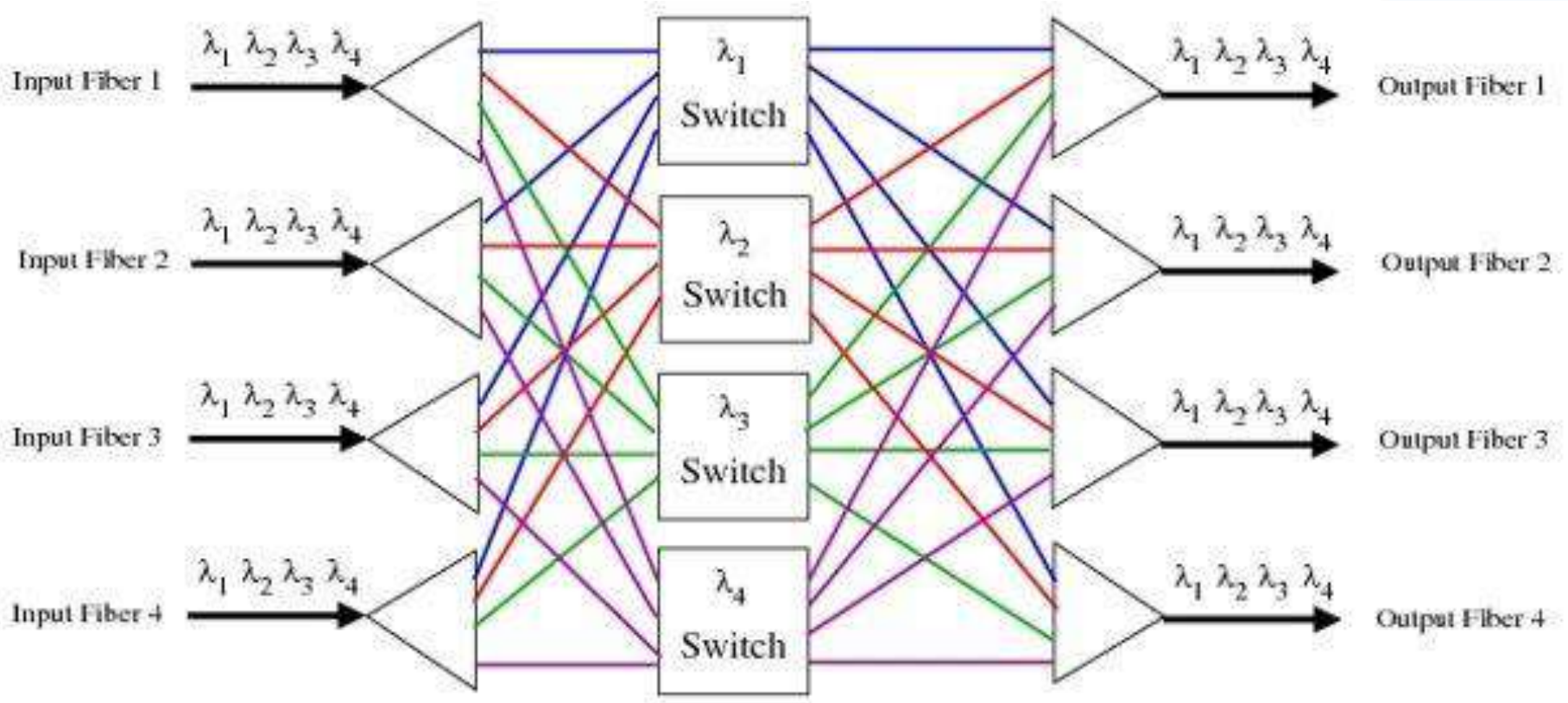
Passive Router *(cont.)*

Note the following about passive routers:

- The connectivity pattern between wavelengths on input fibers and the output fibers is fixed and is defined by the “routing matrix” of the router. The routing matrix is determined by the internal connections between the DEMUX and MUX stages inside the router
- Passive routers are commercially available under several names such as: waveguide grating routers (WGRs), arrayed waveguide grating (AWG), wavelength routers (WRs), and Latin routers.

Active Switch

- Active switches are commonly known as wavelength-routed switch (WRS), wavelength-selective crosseconnect (WSXC).



Active Switch

- The concept of active switches is similar to that of passive routers. However, unlike passive routers in which routing matrix is fixed, active switches have a *reconfigurable* routing matrix.
- The principle of operation of an active switch is simple. Similar wavelengths from all input fibers are connected to the same “space” switching fabric. A given switching fabric can be configured to direct the same wavelength from different fibers to any of the output fibers.

Active Switch *(cont.)*

- Note that, even though active switches provides a routing flexibility for input wavelengths, such switches can be “blocking” under some configurations.
- For instance, it is impossible to configure the active switch in the figure in the previous slide to connect the input signals on the same wavelength to the same output fiber.
- This limitation can be overcome by employing *wavelength converters*.

Optical Interconnection Devices Comparison

Feature	Passive Star	Passive Router	Active Switch
Operation	Passive; does not require power	Passive; relies on optical components	Active; requires external power
Signal Processing	No signal processing; simply splits signals	No signal processing; directs signals based on fixed paths	Performs signal processing; dynamically routes signals
Configuration	Fixed configuration; no dynamic reconfiguration	Fixed paths; limited routing capability	Dynamic configuration; can reconfigure paths in real time
Installation Cost	Generally low due to simplicity	Low; relies on passive components	Higher due to complexity and active components
Scalability	Limited scalability; adds fixed connections only	Limited scalability; designed for specific routing needs	Highly scalable; can manage a larger number of connections dynamically
Applications	Fiber-to-the-home (FTTH), local area networks	WDM systems, fixed routing applications	Data centers, telecommunications, and metro networks
Flexibility	Low; fixed paths limit adaptability	Low; limited to predetermined paths	High; can adapt to changing traffic conditions and reroute dynamically



THANK YOU

QUESTIONS?





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